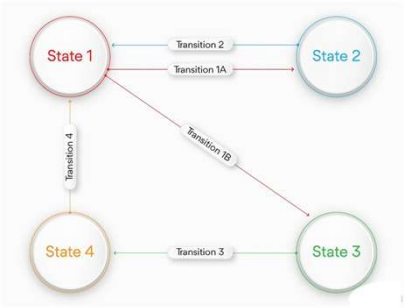


# Chapter 2

# Finite Automata

## Introduction

- The notion of automata **is very old** and was used in the years **1950-1960** for calculation.
  The First processors consisted of a control part and an operative part.
- The control part is an automaton that controls the operative part to perform for example arithmetic and logical operations or to make memory transfers.
- Automata were used in:
  — the design of computer equipment,
  — **word recognition, lexical analysis of a compiler,**
  — text recognition (web form, search engine, etc.),
  — robot programming, planning,
  — property modelling and programme verification,
  — compression of texts,




automata are very old → used a lot in practice nowadays, and still a subject studied in research as problem-solving tool.

# Principles of Automata

- **Question:** Determine tools to recognize if a string belongs to a language
- **Automata:** are mathematical models that take a string of symbols as input and perform a string recognition algorithm.
- **If the algorithm** terminates under certain conditions, The automaton is said to accept the corresponding string.

- Finite Automata **(FA):** known as finite state machines (FSM), are mathematical models with a finite number of states and transitions between those states based on inputs.
- They are commonly used to model systems with a fixed set of behaviors or patterns.

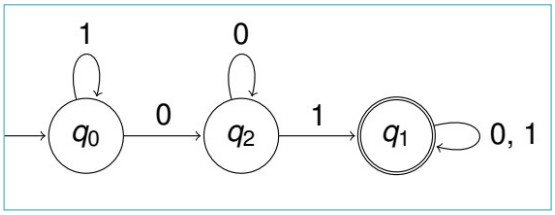
A string $w=a_1a_2...a_n$ is accepted by a FA if there is a **path** in the transition diagram that **starts at the initial state, ends in an acceptor state** and has **a sequence of labels** $a_1a_2...a_n$



# Principles of Automata

## Example

- Set of chains built on {0; 1} **comprising a sequence of characters 01.**
- **3 Scenarios**
  - ✓ The first 0 has not yet appeared
  - ✓ The first 0 has appeared and we are waiting for the first 1 next
  - ✓ Sequence 01 was met



# Principles of Automata




**Automate G** An automata is used to characterize a language (accepted words)

**Definition:**

A finite state **A**utomaton is the five-tune A**= (Q, $V_T$, δ, $q_0$, F)** with

**$V_T$:** a finite alphabet

**Q :** Finite set of states

**F :** All final states

**δ :** Transition function, **δ: Q X $V_T$ → Q**

**$q_0$ :** Initial state



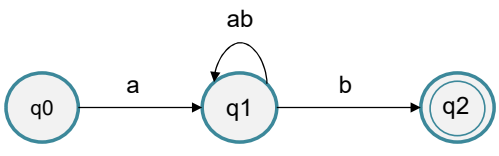
# Principles of Automata

**Application:**

**Let the language L = a (ab)*b**

- **Q** = {q0, q1, q2}
- **$V_T$**= {a,b,ab}
- **q0**=q0
- **δ** = Q x VT→ Q; **δ** (q0, a) =q1; **δ**(q1, b) =q2; **δ**(q1, ab) = q1.

| | a | b | ab |
|---|---|---|---|
| q0 | q1 | - | - |
| q1 | - | q2 | q1 |
| q2 | - | - | - |



# Principles of Automata

- **Example**

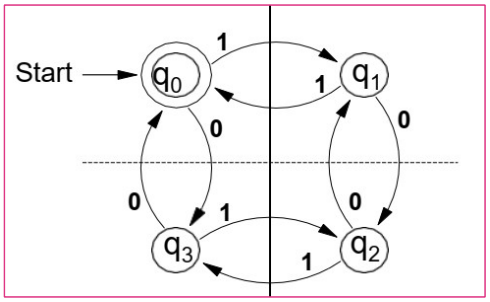
A finite automaton *M* accept all of words on the alphabet *{0,1}* with an even number of **0** and **1**.

**M = ({q0, q1, q2, q3}, {0, 1}, δ, q0, {q0})**

δ

| | 0 | 1 |
|---|---|---|
| $q_0$ | $q_3$ | $q_1$ |
| $q_1$ | $q_2$ | $q_0$ |
| $q_2$ | $q_1$ | $q_3$ |
| $q_3$ | $q_0$ | $q_2$ |

Transition Diagram

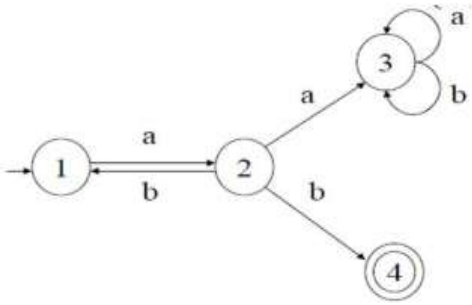





# Principles of Automata

- **Example**

The following transition diagram describes a language whose alphabet is Σ={a, b}.
• The set of states = {1, 2, 3, 4}.
• The final state (acceptor): 4



# Automata from RegEx

- **Method : Glushkov construction**

- Construction of a **FA** accepting the set of words described by a regular expression
- The main steps of the algorithm:

1. Transform the RE to a new expression where each symbol appears at most once in the **RE2**
2. Build the automaton **A'** accepting the set of words described by RE2
3. Transform **A**' into an automaton A accepting the set of words described by the initial RE



# Automata from RegEx

- **Method : Glushkov construction**

1. Transform the RE to a new expression where each symbol appears at most once in the RE2
   - Numbering from 1, the symbols of the expression RE

   **Example: RE = a*b (a|c)* b**

   **RE2 = a1*b2(a3|c4)* b5**

   - Identify the initial state and the set of states of the automaton A' q0=0, **Q**={0,1,2,3,4,5}

# Automata from RegEx

- **Method : Glushkov construction**

2 Build the automaton A' accepting the set of words described by RE2
- Identify the **start and end letters**:

**a**. Find all start letters→start letters ={a1, b2}

**b**. Find all end letters→ end letters ={b5}

**i.** T={end letter numbers}→ F={5}

**ii**.If the empty word can be generated by this expression, add the initial state 0 to F



# Automata from RegEx

- **Method : Glushkov construction**

Find all factors of length 2 of RE2 ($a_i a_j$) :
RE2 = a1*b2(a3|c4)* b5

***The factors are**: a1a1 a1b2 b2a3 b2c4 b2b5 a3a3 a3c4*
*c4a3 c4c4 a3b5 c4c4 c4b5*

Determine the set of transitions of the automaton A'

a. Placing an arc from state **q0** to each state with a leading letter number labeled $\mathbf{a_i}$
b. Place an arc of each state **i** of **ai** to state **j** of aj that has the label a

# Automata from RegEx

- **Method : Glushkov construction**

Determine the set of transitions of the automaton **A'**

a. Placing an arc from state q0 to each state with a start l letter number labeled $a_i$ Transition (0, a1)= 1 Transition (0, b2)= 2

b. Place an arc of each state i of ai to state j of aj that has the label $a_j$

- Transition (1,a1)= 1 Transition (1,b2)=2 Transition (2,a3)=3
- Transition (2, c4)=4 Transition (2,b5)=5 Transition (3,a3)=3
- Transition (3,c4)=4 Transition (4,a3)=3 Transition (4,c4)=4
- Transition (3,b5)=5 Transition (4,b5)=5

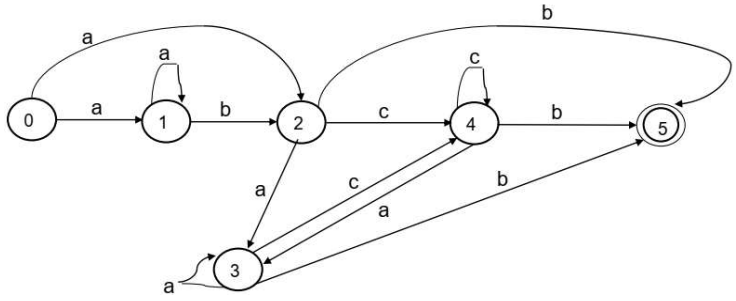


# Automata from RegEx

- **Method : Glushkov construction**

3 Transform A' into an automaton A accepting the set of words described by RE

- Replace the numbered symbols with the original symbols and graphically represent the automaton.



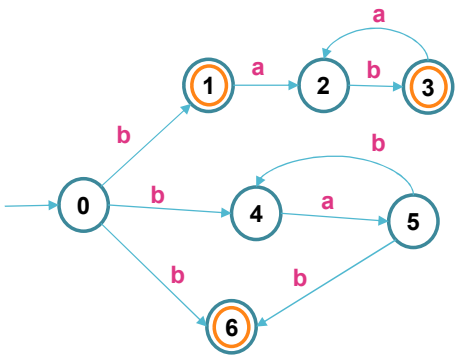
# Automata from RegEx

- Method : Glushkov construction

**Example** b(ab)*|(ba)* b

➔ b1(a2b3)* | (b4a5)*b6

| | successors |
|---|---|
| ϕ | b1 , b4 , b6 |
| b1 | a2 |
| a2 | b3 |
| b3 | a2 |
| b4 | a5 |
| a5 | b4 b6 |
| b6 | ϕ |






# Automata from RegEx

- Method : Glushkov construction

**Example2 :** (C+|a) b (ac)* b

# Automata from RegEx

- **Minimisation of Automata,**

Minimization of **automata** of transforming a given automata into an equivalent one with the smallest possible number of states — that is, one that recognizes the same language but has no redundant states.

**Why minimize?**

- Reduces memory and computation cost
- Easier to understand and visualize
- Used in compiler design, pattern matching, and model checking



# Automata from RegEx

- **Minimisation of Automata,**

1. **Remove Unreachable** StatesFrom the start state, mark all states that can be reached by some input string.Any state not marked is unreachable → remove it.

2. **Identify Distinguishable States :**
   Two states are distinguishable if there exists at least one input string that leads to:
   One state reaching an accepting state, and The other reaching a non-accepting state.

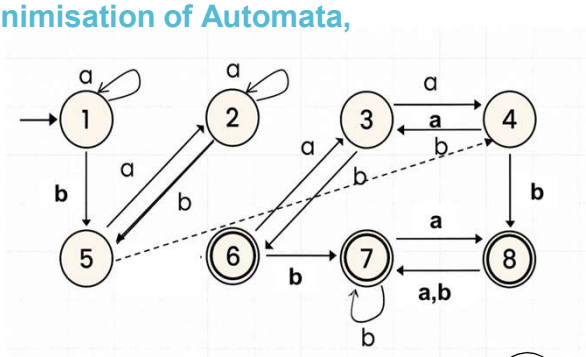
   We find equivalent states (indistinguishable ones) that can be merged.

**The process:**

| Step | |
|---|---|
| 1 | Divide states into two groups: accepting and non-accepting |
| 2 | Check for each pair of states if they behave differently for some input symbol |
| 3 | Split groups further until no more splitting is possible |
| 4 | Merge all equivalent states |

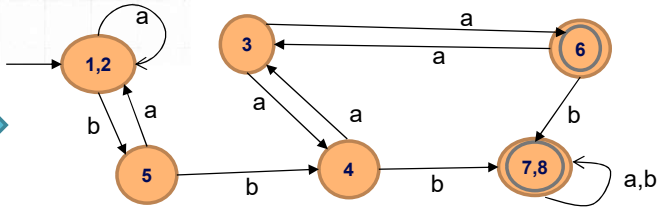
# Automata from RegEx

- Minimisation of Automata,




| | a | b |
|---|---|---|
| 1 | 1 | 5 |
| 2 | 2 | 5 |
| 3 | 4 | 6 |
| 4 | 3 | 8 |
| 5 | 2 | 4 |
| 6 | 3 | 7 |
| 7 | 8 | 7 |
| 8 | 7 | 7 |

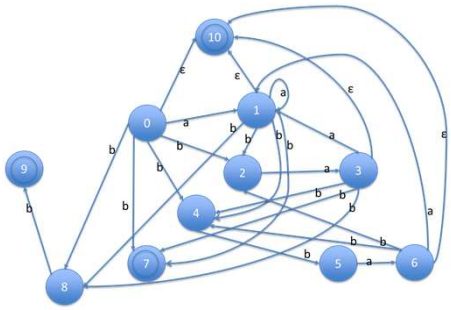
**Step0 : (6,7,8) (1,2,3,4,5)**
**Step1 : (6) (7,8) (1,2,3,4,5)**
**Step2 : (6) (7,8) (3) (4) (1,2,5)**
**Step3: (6) (7,8) (3) (4) (1,2) (5)**

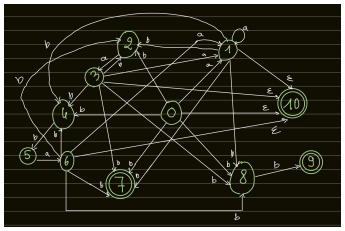



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# Automata from RegEx

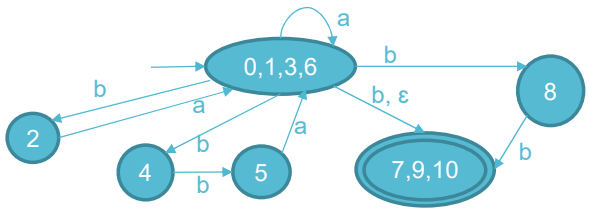
**Exercise : minimisation** (a |ba| bba)* (b|bb|ε)





| | a | b | ε |
|---|---|---|---|
| 0 | 1 | 2,4,7,8 | 10 |
| 1 | 1 | 2,4,7,8 | 10 |
| 2 | 3 | - | - |
| 3 | 1 | 2,4,7,8 | 10 |
| 4 | - | 5 | - |
| 5 | 6 | - | - |
| 6 | 1 | 2,4,7,8 | 10 |
| 7 | - | - | - |
| 8 | - | 9 | - |
| 9 | - | - | - |
| 10 | - | - | - |

Step1: {7,9,10}{0,1,2,3,4,5,6,8}
Step2: {7,9,10}{8}{2,4,5}{0}{1}{3}{6}
Step3: {7,9,10}{8}{2}{4}{5} {0}{1}{3}{6}
Step3: {7,9,10}{8}{2}{4}{5}{0,1,3,6} * fusion (same behaviour)




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